

Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

Summarize your learnings from the lab here.

In this lab we learned how to use a 4 NAND chip to physically wire a logic circuit on a breadboard. We first converted the logic function to NAND only then mapped out how we were going to wire it on the board. We learned how to navigate the breadboards wiring (VCC and GROUND) and use that to connect to our NAND chip, physically wire all the needed cables, and then take that result and output it to a lightbulb.

Lab Questions

1 - Write down DeMorgan's Law and the truth tables proving it out.

$\text{NAND}(A,B) = \text{NOT}(A) \text{ or } \text{NOT}(B)$

$\text{NOR}(A,B) = \text{NOT}(A) \text{ and } \text{NOT}(B)$

A	B	NAND(A,B)	NOT(A) or NOT(B)	NOR(A,B)	NOT(A) and NOT(B)
0	0	1	1	1	1
0	1	1	1	0	0

1	0	1	1	0	0
1	1	0	0	0	0

2 - What is the value in converting circuits to NAND only?

The NAND circuit is known as the universal circuit and that all of the other circuits (AND, NOT, NOR, OR) can be built into a NAND circuit. This prevents design flaws from AND or OR circuits and gives the designer a way to build out a circuit design with only one gate.

3 - How does what you did in lab with the breadboard relate to the FPGA?

With the breadboard we learned how to use VCC and GROUND power to help build a logic circuit, we used physical wiring to complete this. With FPGA you use a software such as verilog to help build the logic out, using the breadboard helps us because we got a more hands-on learning experience and now when we transition to the software side we can appreciate it a little bit more.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.

There is no code for this assignment.